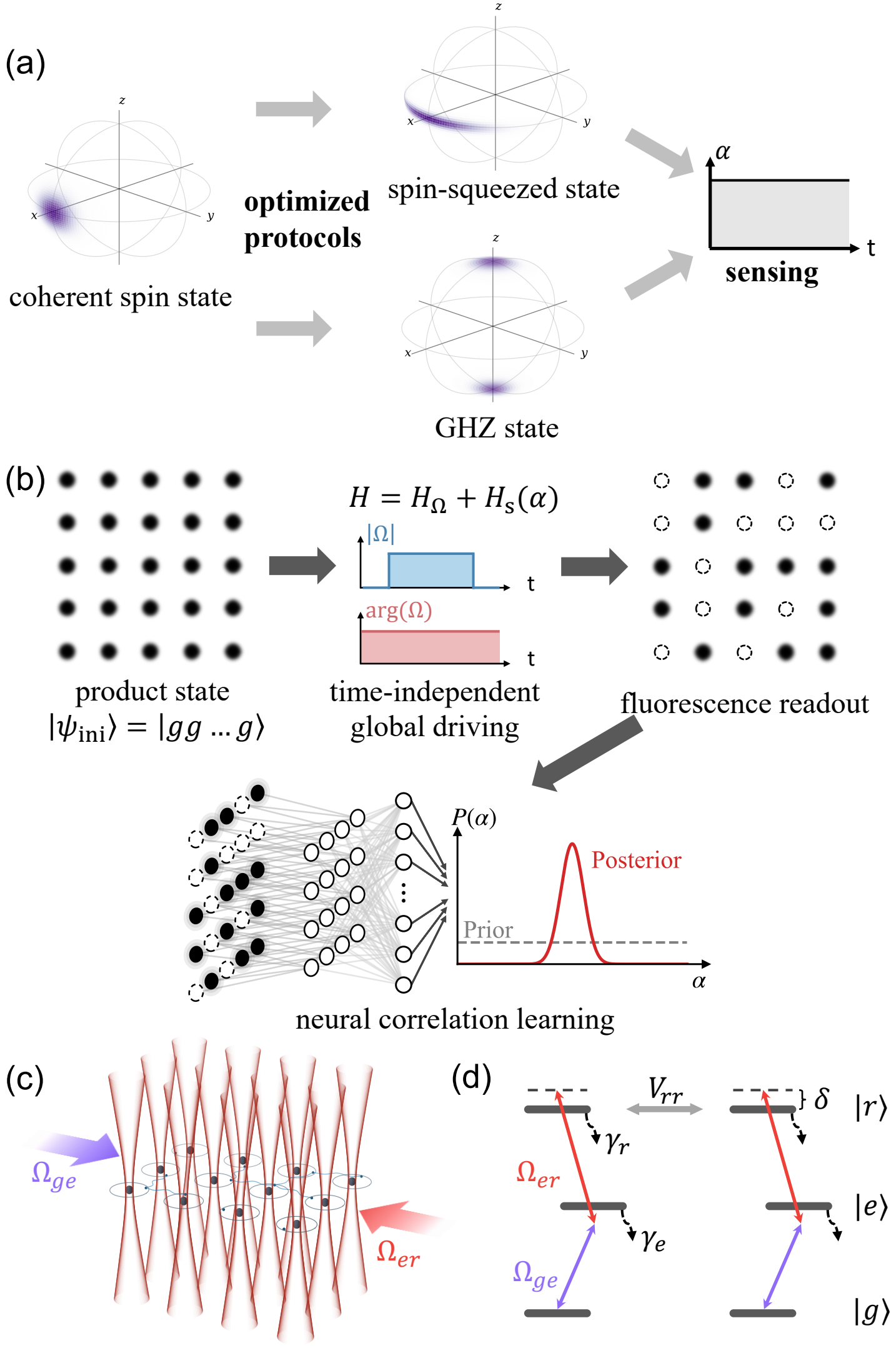

(a)
coherent spin state
optimized protocols
spin-squeezed state
GHZ state
sensing
(b)
product state
$|\psi_{\text{ini}}\rangle = |gg\ldots g\rangle$
$H = H_\Omega + H_s(\alpha)$
$|\Omega|$
$\arg(\Omega)$
time-independent global driving
fluorescence readout
$P(\alpha)$
Posterior
Prior
neural correlation learning
(c)
$\Omega_{ge}$
$\Omega_{er}$
(d)
$V_{rr}$
$\delta$
$\gamma_r$
$\gamma_e$
$|r\rangle$
$|e\rangle$
$|g\rangle$